Numerical Methods for ODEs: Exercise Week 9

Problem 1:

Consider the multi-step method

$$y_{n+2} = y_{n+1} + h\left(\frac{5}{12} f_{n+2} + \frac{2}{3} f_{n+1} - \frac{1}{12} f_n\right). \tag{1}$$

Work out analytically its order from the order conditions and show that the method is zero-stable. Confirm numerically its order of convergence for the test equation

$$y'(t) = \lambda y(t), \; y(0) = 1 \tag{2}$$

with $\lambda < 0$.

Problem 2:

Now apply the method above to the initial value problem

$$y'(t) = 1 + y^2(t), \; y(0) = 1. \tag{3}$$

Write down on paper the implicit equation

$$F(y_{n+2}) = 0 \tag{4}$$

that needs to be solved in every time step and use Python's function `fsolve` from `scipy.optimize` to solve it in your code.
(Link: `https://docs.scipy.org/doc/scipy/reference/generated/scipy.optimize.fsolve.html`)

1. Work out the analytical solution to the problem, then run your code for a range of time steps h and compare. What do you observe?
2. Try to explain the observed behaviour. For $y_n = y_{n+1} = 1$, plot the F from (4) for various values of h. What do you observe with respect to its roots? What happens if you fix $y_n = 1$ and h and vary y_{n+1} instead?
3. Look up Lemma 57 and try to work out the Lipschitz constant L. Does the Lemma help explaining the resulting from point 2?